

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihalan Produk:

Product Description:

Cat No. :

Synonyms

CAS No

Molecular Formula

**Guanidinium thiocyanate**
**Guanidinium thiocyanate**

G/0988/48

Guanidinium isothiocyanate; Thiocyanic acid, compound with Guanidine (1:1)

593-84-0

C2 H6 N4 S

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use

Laboratory chemicals.

Uses advised against

No Information available

**Company**

 Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd  
 Hap Seng Business Park, Lot 01-03, 01-04 Aras 1 Unity Square,  
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 Selangor Darul Ehsan, Malaysia.  
 Main line: +60 3-5525 7888

**Supplier**

E-mail address

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Emergency Telephone Number

Tel: +03-5525 7888

 CHEMTREC Malaysia **1-800-815-308** (Malay)

 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                                             |                     |
|---------------------------------------------|---------------------|
| Acute oral toxicity                         | Category 4 (H302)   |
| Acute dermal toxicity                       | Category 4 (H312)   |
| Acute Inhalation Toxicity - Dusts and Mists | Category 4 (H332)   |
| Skin Corrosion/Irritation                   | Category 1 C (H314) |
| Serious Eye Damage/Eye Irritation           | Category 1 (H318)   |
| Chronic aquatic toxicity                    | Category 3 (H412)   |

**Label Elements**


# SAFETY DATA SHEET

Guanidinium thiocyanate

Revision Date 23-Mar-2025

## Signal Word

**Danger**

### Hazard Statements

H302 + H312 + H332 - Harmful if swallowed, in contact with skin or if inhaled  
H314 - Causes severe skin burns and eye damage  
H412 - Harmful to aquatic life with long lasting effects

### Precautionary Statements

#### Prevention

P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P284 - Wear respiratory protection

#### Response

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor  
P330 - Rinse mouth  
P331 - Do NOT induce vomiting  
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P362 + P364 - Take off contaminated clothing and wash it before reuse

#### Storage

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

#### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

### Other Hazards

EUH032 - Contact with acids liberates very toxic gas  
EUH071 - Corrosive to the respiratory tract  
Toxic to terrestrial vertebrates  
This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component                                      | CAS No   | Weight % |
|------------------------------------------------|----------|----------|
| Thiocyanic acid, compound with guanidine (1:1) | 593-84-0 | >95      |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required. Keep eye wide open while rinsing.

#### Skin Contact

Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Call a physician immediately.

#### Ingestion

Immediate medical attention is required. Do NOT induce vomiting. Drink plenty of water. Never give anything by mouth to an unconscious person.

# SAFETY DATA SHEET

Guanidinium thiocyanate

Revision Date 23-Mar-2025

**Inhalation** Remove to fresh air. If not breathing, give artificial respiration. Call a physician or poison control center immediately. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device.

**Self-Protection of the First Aider** Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Most important symptoms and effects, both acute and delayed**

Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation.

**Indication of any immediate medical attention and special treatment needed**

**Notes to Physician** Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

**Extinguishing media**

**Suitable Extinguishing Media**

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Special hazards arising from the substance or mixture**

The product causes burns of eyes, skin and mucous membranes.

**Hazardous Combustion Products**

Hydrogen cyanide (hydrocyanic acid), Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Nitrogen oxides (NO<sub>x</sub>), Sulfur oxides.

**Advice for fire-fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

**Personal Precautions, Protective Equipment and Emergency Procedures**

Use personal protective equipment as required. Evacuate personnel to safe areas. Avoid contact with skin, eyes or clothing.

**Environmental precautions**

Should not be released into the environment. Do not allow material to contaminate ground water system. Do not flush into surface water or sanitary sewer system.

**Methods and Material for Containment and Cleaning Up**

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

**Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

# SAFETY DATA SHEET

Guanidinium thiocyanate

Revision Date 23-Mar-2025

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe dust. Do not ingest. If swallowed then seek immediate medical assistance.

### Conditions for Safe Storage, Including any Incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Protect from light. Corrosives area. Keep under nitrogen.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

### Exposure Controls

#### **Engineering Measures**

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### **Eye Protection**

Goggles

#### **Hand Protection**

Protective gloves

#### **Skin and body protection**

Long sleeved clothing

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

#### **Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

#### **Recommended Filter type:**

Particulates filter conforming to EN 143

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

When RPE is used a face piece Fit Test should be conducted

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

### Environmental exposure controls

Prevent product from entering drains

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

# SAFETY DATA SHEET

Guanidinium thiocyanate

Revision Date 23-Mar-2025

## Information on basic physical and chemical properties

|                                                |                                 |                                   |
|------------------------------------------------|---------------------------------|-----------------------------------|
| Appearance                                     | Off-white                       |                                   |
| Physical State                                 | Powder Solid                    |                                   |
| Odor                                           | Odorless                        |                                   |
| Odor Threshold                                 | No data available               |                                   |
| pH                                             | approx 4.8 - 6.0                | 20% aq. solution                  |
| Melting Point/Range                            | 118 - 122 °C / 244.4 - 251.6 °F |                                   |
| Softening Point                                | No data available               |                                   |
| Boiling Point/Range                            | No information available        |                                   |
| Flash Point                                    | No information available        | Method - No information available |
| Evaporation Rate                               | Not applicable                  | Solid                             |
| Flammability (solid,gas)                       | No information available        |                                   |
| Explosion Limits                               | No data available               |                                   |
| Vapor Pressure                                 | negligible                      |                                   |
| Vapor Density                                  | Not applicable                  | Solid                             |
| Specific Gravity / Density                     | No data available               |                                   |
| Bulk Density                                   | No data available               |                                   |
| Water Solubility                               | 1420 g/L (20°C)                 |                                   |
| Solubility in other solvents                   | No information available        |                                   |
| Partition Coefficient (n-octanol/water)        |                                 |                                   |
| Component                                      | log Pow                         |                                   |
| Thiocyanic acid, compound with guanidine (1:1) | -1.38                           |                                   |
| Autoignition Temperature                       | Not applicable                  |                                   |
| Decomposition Temperature                      | No data available               |                                   |
| Viscosity                                      | Not applicable                  | Solid                             |
| Explosive Properties                           | No information available        |                                   |
| Oxidizing Properties                           | No information available        |                                   |
| Molecular Formula                              | C2 H6 N4 S                      |                                   |
| Molecular Weight                               | 118.16                          |                                   |

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

Yes. Contact with acids liberates very toxic gas.

### Chemical Stability

Light sensitive.

### Possibility of Hazardous Reactions

|                          |                               |
|--------------------------|-------------------------------|
| Hazardous Polymerization | No information available.     |
| Hazardous Reactions      | None under normal processing. |

# SAFETY DATA SHEET

Guanidinium thiocyanate

Revision Date 23-Mar-2025

## Conditions to Avoid

Avoid dust formation. Incompatible products. Exposure to light. Excess heat.

## Incompatible Materials

Acids. Strong oxidizing agents.

## Hazardous Decomposition Products

Hydrogen cyanide (hydrocyanic acid). Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Nitrogen oxides (NO<sub>x</sub>). Sulfur oxides.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### Product Information

|                     |            |
|---------------------|------------|
| (a) acute toxicity; |            |
| Oral                | Category 4 |
| Dermal              | Category 4 |
| Inhalation          | Category 4 |

| Component                                      | LD50 Oral                          | LD50 Dermal | LC50 Inhalation                                                                                   |
|------------------------------------------------|------------------------------------|-------------|---------------------------------------------------------------------------------------------------|
| Thiocyanic acid, compound with guanidine (1:1) | LD50 = 593 mg/kg (rat)<br>OECD 401 | -           | LC50 combined: 5.319 mg/L<br>LC50 Males: 7.655 mg/L<br>LC50 Females: 3.181 mg/L<br>(Rat) OECD 403 |

(b) skin corrosion/irritation; Category 1 C

(c) serious eye damage/irritation; Category 1

(d) respiratory or skin sensitization;  
    Respiratory Based on available data, the classification criteria are not met  
    Skin Based on available data, the classification criteria are not met

(e) germ cell mutagenicity; Based on available data, the classification criteria are not met

(f) carcinogenicity; Based on available data, the classification criteria are not met  
There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; Based on available data, the classification criteria are not met

(h) STOT-single exposure; Based on available data, the classification criteria are not met

(i) STOT-repeated exposure; Based on available data, the classification criteria are not met  
    Target Organs None known.

# SAFETY DATA SHEET

Guanidinium thiocyanate

Revision Date 23-Mar-2025

(j) aspiration hazard; Not applicable  
Solid

**Symptoms / effects, both acute and delayed** Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation.

**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

**Ecotoxicity effects** Harmful to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component                                      | Freshwater Fish                            | Water Flea         | Freshwater Algae | Microtox |
|------------------------------------------------|--------------------------------------------|--------------------|------------------|----------|
| Thiocyanic acid, compound with guanidine (1:1) | Poecilia reticulata:<br>LC50=89.1 mg/L 96h | EC50=42.4 mg/L 48h |                  |          |

### Persistence and degradability

**Persistence** Persistence is unlikely.

| Component                                                        | Degradability |
|------------------------------------------------------------------|---------------|
| Thiocyanic acid, compound with guanidine (1:1)<br>593-84-0 (>95) | 46% OECD302B  |

**Degradation in sewage treatment plant** Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Bioaccumulative potential** Bioaccumulation is unlikely

| Component                                      | log Pow | Bioconcentration factor (BCF) |
|------------------------------------------------|---------|-------------------------------|
| Thiocyanic acid, compound with guanidine (1:1) | -1.38   | No data available             |

**Mobility in soil** The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils.

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

**Other adverse effects** No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

**Waste from Residues/Unused Products** Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point.

**Other Information** Do not flush to sewer Waste codes should be assigned by the user based on the application for which the product was used Do not empty into drains Large amounts will affect pH and harm aquatic organisms Do not let this chemical enter the environment

# SAFETY DATA SHEET

Guanidinium thiocyanate

Revision Date 23-Mar-2025

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

UN-No UN3261  
Hazard Class 8  
Packing Group III  
Proper Shipping Name Corrosive solid, acidic, organic, n.o.s. Guanidine thiocyanate

### Road and Rail Transport

UN-No UN3261  
Hazard Class 8  
Packing Group III  
Proper Shipping Name Corrosive solid, acidic, organic, n.o.s. Guanidine thiocyanate

### IATA

UN-No UN3261  
Hazard Class 8  
Packing Group III  
Proper Shipping Name Corrosive solid, acidic, organic, n.o.s. Guanidine thiocyanate

Special Precautions for User No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories X = listed

| Component                                      | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL |
|------------------------------------------------|-----------|------|-----|-------|------|------|-------|------|------|
| Thiocyanic acid, compound with guanidine (1:1) | 209-812-1 | X    | X   | X     | X    | X    | X     | X    | -    |

### National Regulations

Persistent Organic Pollutant This product does not contain any known or suspected substance  
Ozone Depletion Potential This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals



# SAFETY DATA SHEET

Guanidinium thiocyanate

Revision Date 23-Mar-2025

**WEL** - Workplace Exposure Limit  
**ACGIH** - American Conference of Governmental Industrial Hygienists  
**RPE** - Respiratory Protective Equipment  
**LC50** - Lethal Concentration 50%  
**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average  
**IARC** - International Agency for Research on Cancer  
**LD50** - Lethal Dose 50%  
**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road  
**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code  
**OECD** - Organisation for Economic Co-operation and Development  
**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association  
**MARPOL** - International Convention for the Prevention of Pollution from Ships  
**ATE** - Acute Toxicity Estimate  
**VOC** - (Volatile Organic Compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>  
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Revision Date** 23-Mar-2025  
**Revision Summary** Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**